🧪 Experiment 1: Communication Noise Analysis

🔷 Definition of Noise

English:
Noise is any unwanted disturbance that affects a signal during transmission.

বাংলা:
Noise হলো এমন একটি অবাঞ্ছিত সংকেত যা মূল signal-এর সাথে মিশে গিয়ে তাকে বিকৃত করে।

🔷 Types of Noise

1. Additive White Gaussian Noise (AWGN)

- Follows **Gaussian distribution**
- Added directly to signal

$$y(t) = x(t) + n(t)$$

বাংলা:
এটি একটি random noise যা signal-এর সাথে সরাসরি যোগ হয় এবং এর মান normal distribution অনুসরণ করে।

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2. Impulse Noise

- Sudden spikes (high amplitude)
- Short duration

বাংলা:
হঠাৎ বড় amplitude এর স্পাইক তৈরি করে, যা signal-এ sharp disturbance আনে।

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3. Thermal Noise

- Due to random motion of electrons

বাংলা:
ইলেকট্রনের random motion এর কারণে হয়, সবসময় উপস্থিত থাকে।

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4. Random Noise

- Completely unpredictable

🔷 Effect of Noise

- Signal distortion
- Error probability increases
- Signal quality decreases

🔷 Signal-to-Noise Ratio (SNR)

$$\text{SNR} = \frac{P_{signal}}{P_{noise}}$$

- High SNR → Better signal
- Low SNR → Poor signal

বাংলা:
SNR বেশি হলে signal পরিষ্কার থাকে, কম হলে noise বেশি থাকে।

🧪 Experiment 2: Random Noise (Mean & Variance)

🔷 Mean (μ)

- Average value of noise

বাংলা:
Noise-এর গড় মান

👉 Mean ≠ 0 → signal shift হয়
(উপর বা নিচে সরে যায়)

🔷 Variance (σ^2)

- Spread of noise

বাংলা:
Noise কতটা ছড়িয়ে আছে

👉 Variance বেশি → বেশি distortion

🔷 Practical Observation (From your lab output)

- Variance বাড়লে SNR কমে
- Mean change হলে signal shift হয়

🔷 Key Concept

- Low variance → smooth signal
- High variance → noisy signal

🧪 Experiment 3: Noiseless Binary Channel

🔷 Definition

A channel where:

- No noise
- No error

🔷 Property

$$\text{Output} = \text{Input}$$

বাংলা:
যা পাঠানো হয়, ঠিক তাই পাওয়া যায়

🔷 From Lab

- Transmitted data = Received data
- Perfect communication

🔷 Capacity

From theory:

- Capacity = 1 bit per transmission

  Elements of IT TM Cover highlet...

🔷 Reality

বাস্তবে এমন channel পাওয়া যায় না

🧪 Experiment 4: Binary Symmetric Channel (BSC)

🔷 Definition

- Binary input (0,1)
- Binary output
- Error probability = p

🔷 Channel Behavior

- Correct transmission → $1 - p$
- Bit flip → p

🔷 Channel Capacity

$$C = 1 - H(p)$$

Where entropy:

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$$

🔷 Explanation

- p = 0 → No error → C = 1
- p = 0.5 → Maximum error → C = 0

🔷 From Lab

- Capacity depends on error probability

🔷 Meaning

English: Maximum reliable data rate
বাংলা: সর্বোচ্চ নির্ভরযোগ্য data transmission rate

🧪 Experiment 5: Shannon–Fano Coding

🔷 Definition

A data compression technique based on probability.

🔷 Basic Idea

- High probability → short code
- Low probability → long code

🔷 Steps

1. Sort symbols by probability
2. Divide into two groups
3. Assign 0 and 1
4. Repeat recursively

🔷 Limitation

- Not always optimal

🔷 From Book Insight

Shannon–Fano is an early coding method but not optimal compared to Huffman

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🔷 বাংলা ব্যাখ্যা

যে symbol বেশি বার আসে, তার code ছোট দেওয়া হয়।

🧪 Experiment 6: Huffman Coding

🔷 Definition

An optimal prefix coding algorithm.

🔷 Key Idea

- Combine lowest probability symbols
- Build binary tree

🔷 Steps

1. Take lowest probability symbols
2. Merge them
3. Repeat
4. Generate tree

🔷 Property

- Minimum average code length
- Optimal compression

🔷 From Book

Huffman coding is proven optimal

Elements of IT TM Cover highlet...

🔷 Comparison

Feature	Shannon-Fano	Huffman
Efficiency	Lower	Higher
Optimal	No	Yes
Method	Top-down	Bottom-up

📊 Important Theory (Very Important for Viva)

🔷 Entropy (H)

$$H = -\sum p(x)\log_2 p(x)$$

English: Measure of uncertainty
বাংলা: অনিশ্চয়তার পরিমাপ

🔷 Channel Capacity

$$C = \max I(X;Y)$$

Meaning:
Maximum reliable transmission rate

Elements of IT TM Cover highlet...

🔷 Communication System Model

- Source → Encoder → Channel → Decoder → Destination

🔷 Analog vs Digital

Analog	Digital
Continuous	Discrete
Noise sensitive	Less sensitive

🔷 SNR Importance

- High SNR → Good quality
- Low SNR → Poor quality

🧠 Final Summary (Exam Style)

English:
Communication systems transmit data through channels where noise affects signals. Channel models like noiseless and BSC help analyze performance. Coding techniques such as Shannon-Fano and Huffman improve efficiency and reduce redundancy.

বাংলা:
Communication system-এ signal পাঠানোর সময় noise signal-কে বিকৃত করে। Channel model (noiseless, BSC) ব্যবহার করে performance বোঝা যায়। Shannon-Fano ও Huffman coding ব্যবহার করে data compression ও efficiency বাড়ানো হয়।

📡 ICT-4102 Lab — **Quick Revision Sheet (1 Page)**

🧪 Experiment 1: Communication Noise

- **Noise:** Unwanted disturbance in signal
- **Types:**
  - AWGN → Gaussian distributed
  - Impulse → Sudden spikes
  - Thermal → Electron motion
  - Random → Unpredictable
- **Effect:** Distortion, errors increase

SNR:

$$\text{SNR} = \frac{P_{signal}}{P_{noise}}$$

👉 High SNR = Good signal

🧪 Experiment 2: Random Noise

- **Mean (μ):** Average value → shifts signal
- **Variance (σ^2):** Spread → distortion level
- **Observation:**
  - ↑ Variance → ↓ SNR
  - Mean ≠ 0 → signal offset

🧪 Experiment 3: Noiseless Binary Channel

- Input = Output
- No error, no noise
- Capacity = **1 bit/use**

- Ideal (not practical)

🧪 Experiment 4: BSC (Binary Symmetric Channel)

- Error probability = **p**
- Correct = $1 - p$, Error = p

Capacity:

$$C = 1 - H(p)$$

Entropy:

$$H(p) = -p\log_2 p - (1-p)\log_2(1-p)$$

👉 p = 0 → C = 1
👉 p = 0.5 → C = 0

🧪 Experiment 5: Shannon–Fano Coding

- Based on probability
- High prob → short code
- Steps:
    a. Sort
    b. Divide
    c. Assign 0/1
- ❌ Not optimal

🧪 Experiment 6: Huffman Coding

- Optimal compression
- Minimum average length
- Steps:
    a. Take lowest probabilities

b. Merge

c. Build tree

- ✅ Optimal

📊 Key Concepts

Entropy:

$$H = -\sum p(x)\log_2 p(x)$$

👉 Measure of uncertainty

Channel Capacity:

$$C = \max I(X;Y)$$

👉 Maximum reliable transmission rate

🔁 Shannon vs Huffman

Feature	Shannon-Fano	Huffman
Optimal	❌	✅
Efficiency	Lower	Higher
Method	Top-down	Bottom-up

🧠 Final Points

- Noise degrades signal quality
- SNR determines performance
- BSC models real errors
- Coding improves efficiency
- Huffman = best compression

🔷 1. Channel Coding vs Source Coding (why coding?)

🔹 Source Coding

English:
Source coding is used to **compress data** by removing redundancy.

বাংলা:
Source coding data compress করার জন্য ব্যবহার করা হয়, অপ্রয়োজনীয় অংশ বাদ দেয়।

👉 Example: Huffman, Shannon-Fano

🔹 Channel Coding

English:
Channel coding is used to **detect and correct errors** during transmission.

বাংলা:
Channel coding transmission-এর সময় error detect ও correct করার জন্য ব্যবহার করা হয়।

👉 Example: Hamming code, parity bit

🔹 Why Coding is Needed?

English:

- To reduce data size (efficiency)
- To improve reliability (error control)

বাংলা:

- Data ছোট করার জন্য
- Error কমানোর জন্য

👉 One-line:
"Source coding → efficiency, Channel coding → reliability"

🔷 2. Analog Signal Types + Range

🔹 What is Analog Signal?

English:
A continuous signal in both time and amplitude.

বাংলা:
সময় ও amplitude উভয় দিকেই continuous signal।

🔹 Types of Analog Signal

👉 Sine wave
👉 Cosine wave
👉 Speech signal
👉 Audio signal

🔹 Audible Range

English:
20 Hz to 20,000 Hz (20 kHz)

বাংলা:
মানুষ ২০ Hz থেকে ২০ kHz পর্যন্ত শুনতে পারে।

🔹 Human Voice Bandwidth

English:
Approximately 300 Hz to 3400 Hz

বাংলা:
মানুষের কণ্ঠস্বর সাধারণত 300–3400 Hz এর মধ্যে থাকে।

🔷 3. Noise Koto Prokar? Effect ki?

🔷 Types of Noise

👉 Gaussian noise
👉 Impulse noise
👉 Thermal noise
👉 White noise

🔷 Effect of Noise

English:

- Signal distortion
- Error increase
- SNR decrease

বাংলা:

- Signal বিকৃত হয়
- Error বাড়ে
- SNR কমে

🔷 4. Entropy–Probability Graph (IMPORTANT)

🔷 Shape

- Curve is **inverted U-shape**

🔷 Key Points

Probability (p)	Entropy
0	0
0.5	Maximum

Probability (p)	Entropy
1	0

🔷 Meaning

English:
Maximum uncertainty occurs when probability is 0.5.

বাংলা:
p = 0.5 হলে অনিশ্চয়তা সবচেয়ে বেশি।

🔷 Viva Line

👉 "Entropy is maximum when both outcomes are equally likely."

🔷 5. Electricity: Voltage, Frequency, Channel Capacity

🔷 Voltage & Frequency (Bangladesh)

- Voltage: **220–240 V**
- Frequency: **50 Hz**

🔷 Channel Capacity of Electrical Line?

👉 Trick question ⚠️

English:
Power line is not designed as a communication channel, so channel capacity is not typically defined like communication systems.

বাংলা:
বিদ্যুৎ লাইন communication channel হিসেবে ডিজাইন করা না, তাই এর নির্দিষ্ট channel capacity সাধারণভাবে নির্ধারিত নয়।

👉 If pushed, say:
Capacity depends on bandwidth and noise (Shannon formula)

🔷 6. 10 Mbps vs 100 kHz

🔹 10 Mbps

English:
Data rate (digital signal)

বাংলা:
এটি data transmission speed (bit per second)

🔹 100 kHz

English:
Frequency of a signal

বাংলা:
এটি signal-এর frequency

🔹 Final Difference

Term	Meaning
Mbps	Data rate
Hz	Frequency

🔹 Viva Line

👉 "Mbps represents how fast data is transmitted, while Hz represents how fast the signal oscillates."

🔷 Bonus (Teacher may cross-question)

❓ Relation between bandwidth and data rate?

Answer:

English:
Higher bandwidth allows higher data rate.

বাংলা:
Bandwidth বেশি হলে data rate বেশি হয়।

🧠 Final Smart Summary

👉

English:
Signals carry information, noise distorts them, coding improves transmission, and capacity defines how much data can be reliably sent.

👉

বাংলা:
Signal তথ্য বহন করে, noise তা বিকৃত করে, coding তা ঠিক করে, আর capacity বলে কত data নির্ভুলভাবে পাঠানো যাবে।